

Master's Comprehensive Examination

Part I

Theory (90 minutes)

December 3, 1994

Instructions: Answer all four questions. Open book and open notes.

1. Let X be a random variable whose density is

$$f_X(x) = \begin{cases} cx^2 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find c .
- (b) Find $P\{\frac{1}{4} \leq X < \frac{3}{4}\}$.
- (c) Find the cdf $F_X(x)$ of X .

2. (a) Jennifer Capriati is returning to the tennis tour. Before she left her percentage of first serves that were good was .6. Assuming that number represents Jennifer's chance of getting a first serve in, what is the probability that she gets at least 100 of her next 200 first serves in? (Assume serves are independent trials.) Answer must be one number to 4 decimal places.

(b) Let X be distributed according to a Poisson distribution with parameter 2. Find the probability that $X \geq 2$ given that $X \geq 1$.

3. Let X_1, X_2, X_3 be a sample of size 3 from a normal distribution with mean 0 and variance σ^2 . Note the density for such a distribution is

$$f_x(x; \sigma^2) = \frac{1}{\sqrt{2\pi} \sigma} e^{-x^2/2\sigma^2}.$$

- (a) Derive the maximum likelihood estimator of σ^2 .
- (b) If the three observations are 1, 2, 3 respectively, what is the maximum likelihood estimate of $(2\sigma^2 - 50)$?

4. (a) Suppose we wish to test $H_0 : \theta = 0$ vs $H_1 : \theta > 0$ for a sample of size n from a normal distribution with mean θ and variance 1. Show that the power function of the UMP test of size α is monotone increasing in θ .

(b) A buyer made 4 measurements of the diameter of buttons in centimeters. The measurements were 19, 21, 13, 15. He was told that the average diameter was 10. Test the hypothesis that the mean of the population is within 5 units of the stated value at the .05 level. (Assume normality and $\sigma^2 = 5$.)

Master's Comprehensive Examination

Part II

Applied (90 minutes)

December 3, 1994

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. A gambler claims that when a nickle is flipped that the probability of getting a head is greater than 0.50. To test this conjecture a nickle is flipped 100 times and 55 heads are observed.

- (a) Construct the appropriate hypotheses.
- (b) Test the hypotheses constructed in part (a), let $\alpha = 0.05$.
- (c) How many heads should be observed to reject H_0 given in part (a)?
- (d) If the probability of a head is really 0.60, what is the probability of rejecting H_0 ?

2. A pharmaceutical company is interested in testing a new drug to relieve headaches. Two clinical trials were conducted.

Trial I

Each patient is given the new drug and the control drug. The patient is instructed to take one of the drugs when a headache occurs and record if the headache has disappeared within one hour. The next time the patient gets a headache the second drug is to be taken and the same outcome is recorded. The headaches are to be at least one week apart and the order the drugs are used is random. The results of the study are given below:

Outcome with control drug	Outcome with new drug	
	Success	Failure
Success	40	5
Failure	15	40

The table shows the number of patients for each possible pair of outcomes. For example, 15 patients had a successful outcome with the new drug and had a failure outcome with the old drug.

Trial II

Each patient is randomly allocated either the new drug or the control drug. The patient is instructed to take the drug when a headache occurs and record if the headache has disappeared within one hour. The results of the study are given below:

	New Drug	Control Drug
Outcome		
Success	55	45
Failure	45	55

- (a) For each trial determine if there is a significant difference between the two drugs with respect to the elimination of a headache within one hour, let $\alpha = 0.05$.

(b) Compare the two trials and make a recommendation regarding which trial you feel is better. Of course, indicate why you have made the recommendation.

3. An experiment was conducted to compare two products with regard to healing time of burns. The experiment was conducted by inflicting a burn on the left and right arms of each subject and then randomizing one of the two products to the left arm and the second product to the right arm. The response recorded was the number of days to healing. Five patients were included in the study and the results are given below:

Subject	Product A	Product B
1	10	8
2	12	11
3	8	6
4	9	6
5	11	10

(a) Using the analysis of variance determine if a significant difference exists between the products, let $\alpha = 0.05$.

(b) Using a paired t-test determine if a significant difference exists between the products, let $\alpha = 0.05$.

(c) Compare the results obtained in parts (a) and (b). Explain your findings.

4. A simple linear regression model

$$Y = \beta_0 + \beta_1 X + \epsilon$$

was fit to a set of data. The following summaries were obtained:

$$\begin{array}{ll} n = 20 & \sum XY = 2000 \\ \sum X = 200 & \sum X^2 = 3000 \\ \sum Y = 100 & \sum Y^2 = 2000 \end{array}$$

(a) Find the least squares estimate of β_0 and β_1 .

(b) Test the hypothesis that $\beta_1 = 0$. Let $\alpha = 0.05$.

(c) Estimate the variance of ϵ .

(d) Obtain a 95% confidence interval for β_1 .

(e) Obtain a 95% confidence interval for $E(Y)$ when $X = 10$.

(f) Find the coefficient of determination. What does this mean?

5. The following hypothesis was tested using a z-test with $\alpha = .05$.

$$H_0 : \mu = 5$$

$$H_1 : \mu \neq 5.$$

The z-test was based on $n = 20$ and $\bar{X} = 3$. From this test H_0 was rejected and H_1 was accepted.

Based on this information determine if each of the following statements is True, False, or there is insufficient information to make a determination. For each statement use:

T for True

F for False

C for cannot be determined.

Also indicate how you have reached your decision.

(a) If $\bar{X}$ and α remain the same and n is increased to 30 H_0 will be rejected and H_1 will be accepted.

(b) If $\bar{X}$ and α remain the same and n is decreased to 10 H_0 will be accepted.

(c) If $\bar{X}$ and n remain the same and α is changed to .10 then H_0 will be accepted.

(d) If $\bar{X}$ and n remain the same and α is changed to .01 then H_0 will be rejected and H_1 will be accepted.

(e) If n and α remain the same and $\bar{X} = 2$ then H_0 will be accepted.

(f) If n and α remain the same and $\bar{X} = 4$ then H_0 will be accepted.

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Master's Comprehensive Examination

Part I

Theory (90 minutes)

March 26, 1994

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Suppose n records are manufactured by a particular company. Let X = number among the n that are warped, $p = P$ (warped record).

a. Derive the maximum likelihood estimator of p . If $n = 20$, $X = 3$, what is the estimate?

b. Is the estimator of (a) unbiased?

c. If $n = 20$, $x = 3$, what is the m.l.e. of $(1 - p)^4$?

2. Let $X_1, \dots, X_n$ be observations from a normal distribution with mean μ and variance 1.

a. Find a 95% confidence interval for μ .

b. If n observations are taken, find n large enough so that the length of the confidence interval is .1.

3. Let X be a normal random variable with mean μ and variance 1. Test $H_0 : \mu = 0$ vs $H_1 : \mu > 0$.

a. Indicate the uniformly most powerful test of size .05.

b. Find the power of this test when $\mu = 1$.

4. A card is drawn from a deck of playing cards and then another is drawn at random from those that are left. Find the probability of the events:

a. Neither are hearts.

b. First is heart, second is not heart.

c. Second is a heart.

5. Let X be a random variable whose density is

$$f_X(x) = \begin{cases} cx^2, & \text{if } 0 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$$

a. Find c .

b. Find $P\{\frac{1}{4} < X < \frac{3}{4}\}$.

c. Find the cdf of X .

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 26, 1994

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. An experiment was conducted to determine the effectiveness of a new drug in treating hypertension. Ten patients were included in the study. Prior to treatment, the systolic blood pressure was determined for each patient. The patient was administered the new drug for a six-week period. At the completion of the study the systolic blood pressure was measured again. The data from this study are given below:

Patient	Systolic Blood Pressure	
	Baseline	Final
1	160	140
2	180	170
3	150	160
4	150	130
5	170	170
6	180	150
7	200	180
8	140	160
9	190	150
10	180	160

a) Determine if the drug has resulted in a significant *reduction* from baseline in systolic blood pressure. In selecting a statistical method you may assume the data are normally distributed. Be sure to show the null and alternative hypotheses.

b) When *designing* this study the experimenter made one important mistake. Indicate what change you feel is necessary to make this study acceptable to careful review.

2. A calibration curve was developed prior to using an instrument to determine plasma cholesterol. This process consisted of taking five samples with known cholesterol levels (X) and obtaining an instrument reading (Y) for each sample. The result of one such calibration process is given below:

X	Y
100	15.1
150	19.8
200	25.0
250	30.4
300	34.8

Assuming a calibration model of the form

$$Y = \beta_0 + \beta_1 X + \epsilon,$$

answer the following questions:

- a) Estimate β_0 and β_1 .
- b) Test the hypothesis

$$H_0 : \beta_1 = 0, \quad H_1 : \beta_1 \neq 0.$$

- c) Construct a 95% confidence interval for β .
- d) Two "unknown" plasma samples are processed on the instrument and give readings of 22.5 and 40.0. For each, estimate the cholesterol level.
- e) In (d), which predicted cholesterol level do you believe is better? Why?

3. A three factor factorial experiment was conducted to study the growth of rabbits. The factors included in the experiment were:

Gender – Male, Female

Protein in Diet – Low, Medium, High

Fur Color – Red, White, Black

Four rabbits were placed into each of the 18 treatment combinations. The response variable was the weight gain over a two-month period. A partially completed ANOVA table is given below.

a) Complete the ANOVA table assuming the three factors being studied are fixed effects.

Sources of Variation	Degrees of Freedom	Sum of Squares	Mean Square	Expected Mean Square	<i>F</i>
Total		1500			
Gender		500			
Protein		200			
$G \times P$		50			
Fur		100			
$G \times F$		50			
$P \times F$		30			
$G \times P \times F$		30			
Rabbit ($G \times P \times F$)		540			

b) An analysis of the residuals is also performed. From the analysis it appears that the mean and the standard deviation of the data have a proportional relationship. (Fat bunnies have more variability in their body weight than thin bunnies.) What would you recommend next in the analysis of these data?

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4. To investigate the effect of massive ingestion of ascorbic acid upon food consumption on the proximate body composition of young pigs. Male pigs were force-fed ascorbic acid of 1000mg daily for nine weeks; a control group was included. In Table 1 we present the proximate body compositions (in terms of percent of body weight) of crude lipid in the carcasses of the pigs.

Table 1. Crude Lipid Compositions (%)	
Control Group	1000mg Group
23.8	13.8
15.4	9.3
21.7	17.2
18.0	15.1

a) Test the hypothesis that the mean of the control population equals the mean of the 1000mg population. (Do not assume the data are Normally distributed.) Compute the P -value.

b) Determine a 94.2% nonparametric confidence interval for the difference between the two mean populations.

5. In reading Pap smears, pathologists have difficulty in agreeing on whether a certain minor atypical cellular appearance is abnormal or not. Two pathologists classified the same 52 slides independently, grading them zero, indicating normal, to 4, indicating invasive carcinoma. The results are as follows.

		Pathologist A					
		0	1	2	3	4	
Pathologist B	0	21	3	1	0	0	25
	1	1	4	2	1	0	8
	2	0	2	6	4	0	12
	3	0	0	2	2	0	4
	4	0	0	0	0	3	3
		22	9	11	7	3	52

Is there evidence that one pathologist tends to read more pathology than the other? (Test the hypothesis that neither sees more disease than the other. Compute the P -value.)